



Material Declaration Sheet

Vishay Inter-technology

Date 7/24/2024

Part / Product Family Details

Part / Family Series	RoHS Compliance Status	RoHS Compliance Date Code dd-mm-yyyy	Total product Weight (mg)	Value Range	3rd Party Lab ICP Test Report Available	Manufacturing Location	Number of Exemptions Used
CRCW0402 e3 (10R to 10M)	YES	16-Oct-2004	0.62	10R to 10M	Yes	Israel	1

Technical Information

Terminal Plating / Grid Array Material	Terminal Base Alloy	JESD-97 Pb-Free Material Code Marking	J-STD-20D MSL Rating	Reflow Peak Process Body Temperature	Reflow Maximum number of cycles	Reflow Max.Time at Peak Temperature (sec)	Soldering Compatibility (SnPb/Pb-Free)
Tin (Sn)	Other	e3	1	260 ° C	3	40	Backward & Forward

Material Composition

Homogenous Material Name	Material Classification	Substance Name	CAS number	Weight of Substance (mg)	With respect to Homogenous Material		% with respect to Total Product Weight	RoHS Exemptions Used
					%	ppm		
Bottom Conductor	Ceramics / glass			0.00641				
		Glass, oxide	65997-17-3	0.00022	3.43	34321	0.04	0 - None
		Silver	7440-22-4	0.00619	96.57	965679	1.00	0 - None
Ceramic Substrate	Ceramics / glass			0.50931				
		Aluminium oxide	1344-28-1	0.48894	96.00	960005	78.86	0 - None
		Calcium oxide	1305-78-8	0.00102	0.20	2003	0.16	0 - None
		Magnesium oxide	1309-48-4	0.00407	0.80	7991	0.66	0 - None
		Silicon dioxide	14808-60-7	0.01528	3.00	30001	2.46	0 - None
Front side	Other special metals			0.00028				
		Chromium	7440-47-3	0.00009	32.14	321429	0.01	0 - None
		Copper	7440-50-8	0.00004	14.29	142857	0.01	0 - None
		Nickel	7440-02-0	0.00015	53.57	535714	0.02	0 - None
Nickel plating	Nickel alloys			0.03314				
		Misc., not to declare	System	0.00003	0.10	1000	0.01	0 - None
		Nickel	7440-02-0	0.03311	99.90	999000	5.34	0 - None
Overcoat	Lacquers			0.01300				
		Carbon black	1333-86-4	0.00039	3.00	30000	0.06	0 - None
		Epoxy resin	25068-38-6	0.00936	72.00	720000	1.51	0 - None
		Silicon dioxide	60676-86-0	0.00325	25.00	250000	0.52	0 - None
Passivation	Ceramics / glass			0.00373				
		Lead silicate [Glass]	65997-18-4	0.00373	99.90	999000	0.60	7(c)-I
		Misc., not to declare	System	0.00000	0.10	1000	0.00	0 - None
Resistive element	Ceramics / glass			0.00744				
		Lead silicate [Glass]	65997-18-4	0.00223	29.97	299731	0.36	7(c)-I
		Palladium	7440-05-3	0.00056	7.53	75269	0.09	0 - None
		Ruthenium oxide	12036-10-1	0.00216	29.03	290323	0.35	0 - None
		Silver	7440-22-4	0.00249	33.47	334677	0.40	0 - None
Tin plating	Other special metals			0.03314				
		Misc., not to declare	System	0.00003	0.10	1000	0.01	0 - None
		Tin	7440-31-5	0.03311	99.90	999000	5.34	0 - None
Top Conductor	Ceramics / glass			0.01355				
		Glass, oxide	65997-17-3	0.00027	1.99	19926	0.04	0 - None
		Palladium	7440-05-3	0.00007	0.52	5166	0.01	0 - None
		Silver	7440-22-4	0.01321	97.49	974908	2.13	0 - None

EU-RoHS Directive-2011/65/EU MCV of 0.1% by mass (1000 PPM) in homogeneous material for: Lead (Pb), Mercury, Hexavalent Chromium, Polybrominated Biphenyls (PBB), Polybrominated Diphenyl Ethers (PBDE) and MCV of 0.01% by mass cadmium

Exemption Used 7(c)-I - Electrical and electronic components containing lead in a glass or ceramic other than dielectric ceramic in capacitors, e.g. piezoelectronic devices, or in a glass or ceramic matrix compound

Note :- (i) All information is based on data received from our vendors & subjected to change without prior notice.
(ii) Substance weight are derived from MSDS.

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